Shared memory analysis using weighted vector addition

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1 Observations

The following observations have been made using block size of 32 x 32 and vector size of 10000 (N=100 so N*N=10000)

ncoeff	**Global Memory (μs)**	**Shared Memory (μs)**	**Speedup**
10	6723	188	35.6
100	7364	188	39.1
1000	7862	344	22.9
10000	8309	1652	5.0

Table 1: Comparison of execution times using global and shared memory for vector addition.

1. **Shared memory has limited size:**
   When ncoeff was 10^5 shared memory vector addition showed error which lead to finding that shared memory size is around 48Kb (depends on the gpu but aroudn that range)
2. **Shared memory is at warp level**
   Haven't been able to fully exploit this and hence unable to explain the speedup pattern